Ho Chi Minh City University of Technology
Faculty of Computer Science and Engineering

Electronic Commerce Assignment
Project Report

CampusCart
Student-Centered E-Commerce Platform

Course: CO3027 Electronic Commerce

Lecturer: NGUYN TH ÁI THO

Group Name: INDOMIE

Members:

- Surya Hanjaya
- Arsyad Ghudzamir Afif
- Minh Bùi Trng

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1 Introduction

CampusCart is a student-focused e-commerce platform designed to address the inefficiencies of general marketplaces within a campus environment. The platform targets high-frequency, low-value transactions by offering a streamlined purchasing experience, localized supply, and faster delivery. CampusCart operates under a Business-to-Consumer (B2C) model, where local vendors provide products directly to student users through a centralized platform.

The core problem lies in the mismatch between existing e-commerce systems and student behavior. Current platforms are optimized for broad markets, resulting in longer delivery times, higher relative costs for small orders, and low personalization. These factors reduce the likelihood of frequent purchases among students.

CampusCart addresses this gap through three key mechanisms. First, a pre-order system aggregates demand, allowing vendors to align supply with actual needs and reduce waste. Second, a localized vendor network enables faster delivery, often within the same day. Third, recommendation features improve product discovery, reducing the effort required to complete transactions.

From a business model perspective, CampusCart operates as a volume-driven marketplace. Revenue is generated through a 10% transaction commission and a delivery service margin. Each order produces 7,000 VND in revenue, consisting of 5,000 VND from commission and 2,000 VND from delivery margin.

After accounting for variable costs, each transaction contributes approximately 1,000 VND toward fixed costs. With an estimated monthly fixed cost of 20 million VND, the platform requires 20,000 orders per month to reach break-even.

Early-stage projections indicate monthly transaction volumes between 900 and 2,400 orders. This results in an expected operational deficit, which is consistent with a validation phase focused on user acquisition rather than immediate profitability.

To bridge the gap toward sustainability, CampusCart adopts a multi-campus expansion strategy. Each campus functions as a localized demand cluster, allowing transaction volume to scale without proportional increases in fixed costs. As the platform expands to 8–10 campuses, total order volume approaches the break-even threshold.

In parallel, margin per transaction improves over time as delivery efficiency increases and additional revenue streams are introduced. Delivery efficiency increases through batch processing, reducing cost per delivery. Additional revenue streams, such as promoted listings and vendor subscriptions, further enhance profitability without increasing user prices.

1.1 Problem Statement

At present, students rely on a combination of general e-commerce platforms and local stores to fulfill their daily needs. While these options provide access to a wide range of products, they are not specifically designed for student usage patterns.

As a result, several inefficiencies arise. Prices may vary across platforms, delivery costs can be disproportionately high for small orders, and waiting times often extend from

two to five days. In addition, the lack of tailored recommendations makes it difficult for users to quickly identify relevant products.

These conditions discourage users from completing small or frequent purchases, as the effort required outweighs the perceived value of the transaction. These limitations reduce overall convenience and negatively affect the user experience. Consequently, there is a clear need for a more focused and optimized solution that better aligns with student behavior.

1.2 Proposed Solution

CampusCart addresses these challenges by offering a curated platform specifically tailored to student needs. The system incorporates a pre-order model that aggregates demand, enabling vendors to reduce overproduction, align supply with actual demand, and offer more competitive pricing.

Furthermore, by leveraging a network of local vendors, the platform is able to significantly reduce delivery times, often achieving same-day or next-day fulfillment. In parallel, the integration of AI-based recommendation features suggests products based on user behavior, reducing the number of search interactions required to find relevant products. This shifts the process from manual exploration to guided selection.

Together, these elements create a more efficient and user-friendly ecosystem, delivering lower costs, faster service, and improved customer satisfaction. This combination positions CampusCart as a practical and scalable solution within the student market.

2 Market Analysis

2.1 Target Customers

CampusCart primarily targets university students aged 18–25 who live in or around campus areas. This group represents a highly active digital segment, with strong familiarity and engagement in mobile-based platforms.

In addition to their high level of online activity, students tend to make frequent purchases, often placing multiple small orders throughout the week. These transactions are typically driven by daily needs such as food, study materials, and basic necessities.

Such behavioral patterns create a consistent and predictable demand, making students an ideal customer segment for a specialized e-commerce platform. Their purchasing frequency and reliance on convenience-based services contribute to high platform usage and long-term engagement potential.

2.2 Market Research

The Vietnamese e-commerce market has experienced rapid growth in recent years, reaching an estimated value of approximately $20 billion in 2024. This growth is expected

to continue at an annual rate of 15–20%, reflecting the increasing adoption of online shopping across the country.

A significant portion of this activity is driven by mobile devices, which account for more than 70% of total transactions. This trend highlights the importance of mobile-first platforms, particularly for younger user groups such as students.

Within this expanding market, students stand out as a high-frequency user segment. Their regular purchasing behavior provides a stable revenue base, while their preference for convenience reinforces the importance of accessibility and speed.

These insights suggest that a localized, mobile-optimized platform such as CampusCart is well-positioned to capture demand and benefit from ongoing market growth. This behavior increases transaction frequency per user, which directly supports revenue stability in platform-based business models.

2.3 Competitor Analysis

Platform	Target	Price	Delivery	Personalization
Shopee	Mass market	Medium-low	2–5 days	Low
Lazada	Mass market	Medium	2–5 days	Low
Local Shops	Nearby users	Low	Immediate	None
CampusCart	Students	**Low**	**Fast**	**High**

While major platforms such as Shopee and Lazada benefit from large-scale operations and extensive product offerings, they are not specifically optimized for student needs. Their broad focus often results in lower personalization, leading to higher relative delivery costs and longer decision cycles for small, frequent purchases.

In contrast, CampusCart adopts a niche strategy that prioritizes relevance, speed, and cost efficiency. By focusing exclusively on students and leveraging local vendor networks, the platform is able to deliver faster service and more targeted offerings.

This differentiation enables CampusCart to compete effectively despite its smaller scale, particularly in scenarios where convenience and personalization are more important than product variety.

3 Business Analysis

3.1 Value Proposition

CampusCart delivers an affordable, fast, and personalized shopping experience tailored specifically for university students. This value proposition directly reflects the core characteristics of student purchasing behavior, which prioritizes low cost and minimal effort in completing transactions.

Cost efficiency is achieved through a pre-order mechanism that aggregates demand before order processing. By aligning supply with actual demand, vendors can reduce

excess inventory and avoid overproduction. This allows products to be offered at more competitive prices compared to traditional on-demand models.

In terms of logistics, the platform leverages a network of local vendors located near campus areas. This significantly shortens delivery distances and enables faster fulfillment, often within the same day. Compared to conventional e-commerce delivery cycles, this reduces waiting time from several days to a matter of hours.

To support product discovery, CampusCart integrates AI-based recommendation features that analyze user behavior and past transactions. Instead of requiring users to manually search through large product lists, the system prioritizes relevant items, reducing the number of steps needed to find desired products. This reduces decision time and lowers friction in repeat transactions, making the overall shopping process more efficient.

3.2 SWOT Analysis

Strengths - Niche student focus - Lower cost via pre-order - Fast local delivery - High personalization	**Weaknesses** - Limited initial scale - Dependence on local vendors - Requires user adoption early - Logistics complexity
Opportunities - Growing e-commerce market - High student demand - Mobile-first behavior - Expansion to multiple campuses	**Threats** - Strong competitors (Shopee, Lazada) - Price competition - Vendor reliability risk - Customer switching behavior

3.3 Innovation Aspects

CampusCart introduces an integrated approach that combines a pre-order system, AI-based recommendations, and a smart batch delivery mechanism to improve both user experience and operational efficiency.

The pre-order model allows the platform to aggregate demand before processing orders, aligning supply directly with aggregated demand. In parallel, AI-driven recommendations help users quickly identify relevant products based on their preferences and past behavior, minimizing browsing time and increasing user interaction through faster and more relevant product discovery.

A key innovation lies in the smart batch delivery system. Instead of processing each order individually, the platform groups orders based on location and time windows, allowing deliveries to be consolidated. This reduces redundant delivery routes by grouping

nearby orders into a single batch.

By optimizing delivery operations in this way, CampusCart is expected to gradually reduce delivery costs over time. In the early stage, the average delivery cost is maintained at 8,000 VND per order, consistent with the current revenue model. As order density increases, batch delivery can improve efficiency and increase delivery margin in later stages.

4 Business Model Canvas

The Business Model Canvas presented in Figure 1 illustrates how CampusCart creates, delivers, and captures value within a student-focused e-commerce environment. It provides a structured overview of the key components that define the platform's operations and growth strategy.

At its core, the model is built around university students, a segment characterized by frequent purchasing behavior and strong sensitivity to price and convenience. The platform responds to these needs by offering a combination of affordable pricing, fast delivery, and personalized product recommendations.

Operationally, the system relies on a combination of platform development, vendor management, and logistics coordination. These activities are supported by key resources such as the technology infrastructure, user data, and a network of local vendors. Together, they enable efficient service delivery and scalability.

Revenue is generated primarily through transaction commissions and delivery fees, both of which scale proportionally with user activity. This ensures that growth in platform usage directly translates into increased revenue.

The alignment between customer needs, operational processes, and revenue streams demonstrates the viability of the business model. This structure allows revenue to scale with user activity without requiring proportional increases in operational costs.

5 Feasibility Analysis

5.1 Technical Feasibility

CampusCart can be effectively developed using modern web technologies such as React for the frontend and Node.js for the backend. These technologies are widely adopted in industry and provide strong support for building scalable and maintainable applications.

React enables the development of fast and responsive user interfaces, ensuring a smooth user experience across devices. At the same time, Node.js supports efficient backend processing and seamless API integration, allowing the system to handle real-time interactions and data flow effectively.

In addition, the use of cloud-based infrastructure provides flexibility in scaling resources according to user demand. As the platform grows, the system can be expanded

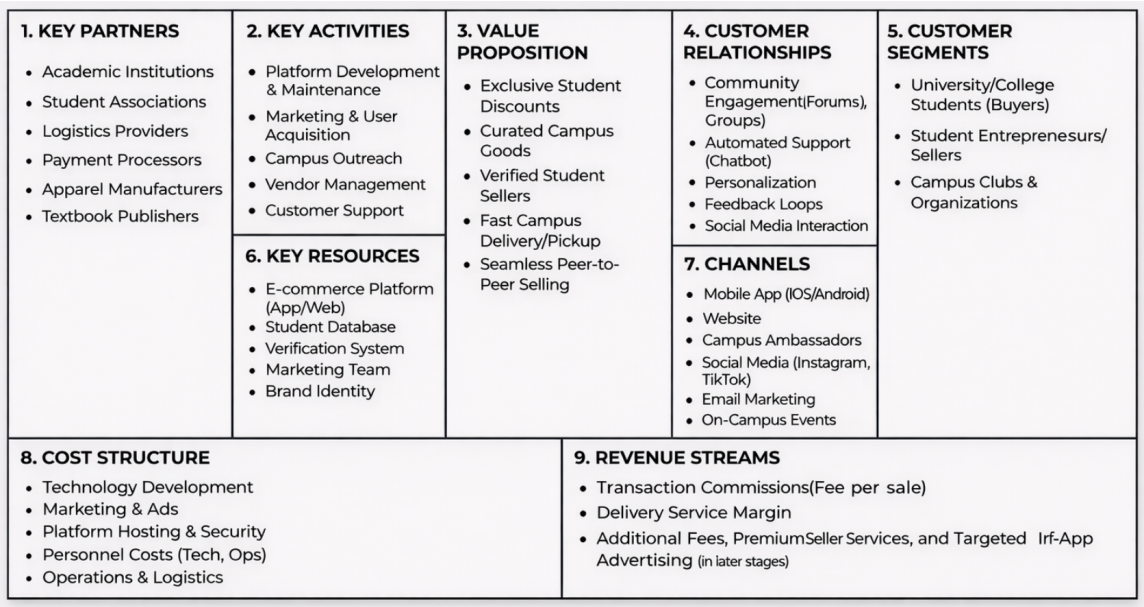

Figure 1: Business Model Canvas illustrating how CampusCart creates and captures value in a student-focused market

without significant changes to its core architecture.

This technology stack supports rapid development, reliable performance, and long-term scalability.

5.2 Financial Feasibility

From a financial perspective, CampusCart requires a relatively low initial investment compared to traditional e-commerce models. This is largely due to the adoption of a pre-order system, which eliminates the need for inventory storage and reduces upfront product costs.

By aggregating demand before processing orders, the platform avoids warehouse expenses and minimizes the risk of unsold inventory. Furthermore, collaboration with local vendors helps reduce operational and logistics costs while maintaining service efficiency.

This cost-efficient structure allows the business to scale gradually without requiring heavy infrastructure investment. However, reaching break-even still depends on achieving sufficient transaction volume, as outlined in the financial analysis section.

6 Revenue Model

The purpose of this section is to explain how CampusCart generates income at the transaction level and how this revenue structure supports scalability.

CampusCart operates on a dual-source revenue model consisting of transaction commissions and delivery service margins. Each completed order contributes to platform revenue through these two components.

First, the platform applies a 10% commission on the average order value of 50,000 VND. This results in 5,000 VND revenue per order. This commission represents the platform's core monetization mechanism and scales directly with transaction value.

Second, users are charged a delivery fee of 10,000 VND per order. The actual delivery cost is estimated at 8,000 VND. The difference between these values produces a net delivery margin of 2,000 VND per order.

Combining both components, the total revenue per transaction is calculated as follows:

$$\text{Total Revenue per Order} = 5,000 + 2,000 = 7,000 \text{ VND}$$

This structure ensures that each transaction contributes positively to revenue while maintaining price levels that remain acceptable for students.

The overall flow of revenue generation is illustrated in Figure 2. The diagram shows how user payments are distributed between vendors, logistics, and the platform.

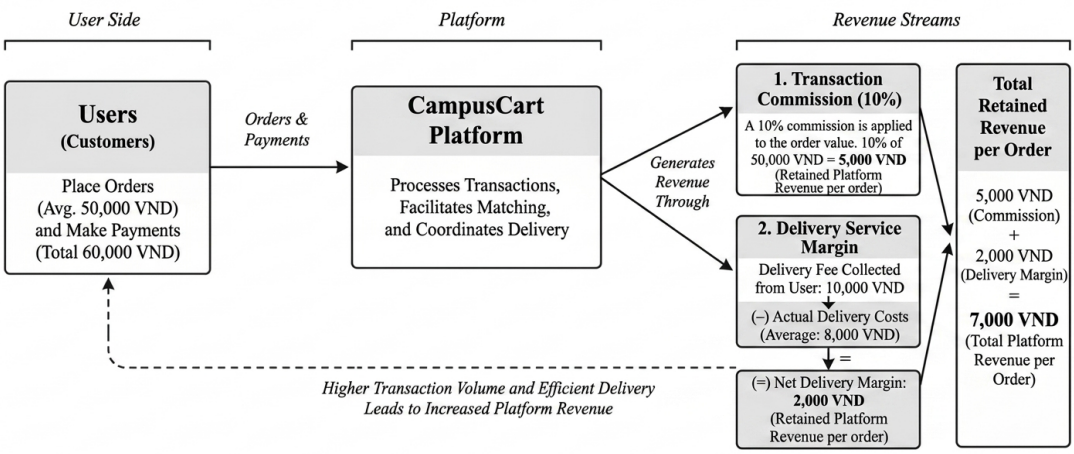

Figure 2: Revenue model illustrating how CampusCart generates income through transaction commissions and delivery margins

From a scaling perspective, the model follows a linear pattern. As transaction volume increases, total revenue grows proportionally. For example, 1,000 orders generate 7 million VND in revenue, while 10,000 orders generate 70 million VND.

However, revenue alone does not determine profitability. Each transaction incurs variable costs, which must be considered to understand the actual contribution of each order.

7 Unit Economics Analysis

To evaluate financial sustainability, it is necessary to break down the cost structure at the transaction level.

Unit economics describes how much value each order contributes after accounting for direct costs. This provides a more accurate view of business performance than revenue alone.

Table 1: Unit Economics of CampusCart

Metric	Value	Notes
Average Order Value	50,000 VND	Typical student spending
Commission per Order	5,000 VND	10% platform fee
Net Delivery Margin	2,000 VND	After delivery cost
Total Revenue per Order	7,000 VND	Platform revenue
Payment Processing Fee	1,500 VND	Payment gateway cost
Operational Handling Cost	2,000 VND	Customer support and system operations
Promotional Subsidy	2,500 VND	Discounts and incentives
Total Variable Cost	6,000 VND	Per order cost
Contribution Margin	1,000 VND	Net contribution per order

From this breakdown, each order contributes 1,000 VND after covering variable costs.

This contribution margin is relatively small. As a result, the business depends on high transaction volume to cover fixed costs. In practical terms, the model behaves similarly to a high-volume retail system where profitability emerges from scale rather than high margins per unit.

This structure also introduces sensitivity to cost changes. For instance, a 10% increase in delivery cost or promotional spending would significantly reduce the contribution margin. Therefore, maintaining operational efficiency is critical from the early stages.

8 Pricing Strategy and Break-Even Analysis

This section evaluates how the current pricing structure translates into overall financial viability.

CampusCart adopts a penetration pricing approach during the early stage. The objective is to reduce barriers to entry and encourage frequent usage among students. Lower margins are accepted initially in exchange for faster user acquisition and higher transaction volume.

To determine when the business becomes financially sustainable, the break-even point (BEP) is calculated.

The BEP represents the number of orders required to cover all fixed costs using the contribution margin generated per transaction.

Given:

- Contribution margin per order = 1,000 VND

- Fixed monthly cost = 20,000,000 VND

The break-even point is:

$$BEP = \frac{20,000,000}{1,000} = 20,000 \text{ orders/month}$$

This is equivalent to approximately 667 orders per day.

Figure 3 illustrates the relationship between total cost and total revenue, and where they intersect.

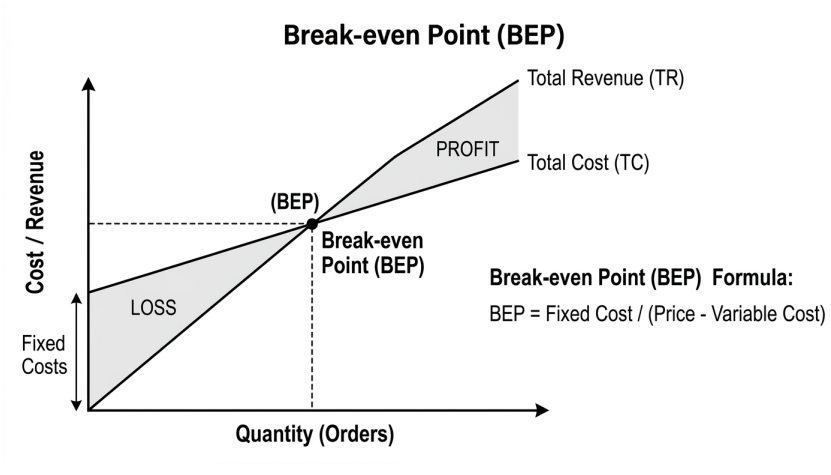

Figure 3: Break-even analysis showing the intersection of total cost and total revenue

When compared with early-stage projections of 900 to 2,400 orders per month, the gap becomes clear. The platform initially operates at only 4.5% to 12% of the required break-even volume.

This implies that profitability cannot be achieved within a single-campus setup. Instead, growth must follow a multi-campus expansion strategy.

A realistic scaling path is as follows:

- 1 campus: demand validation and user acquisition
- 3–5 campuses: increasing order density and partial cost coverage
- 8–10 campuses: approaching break-even
- 10+ campuses: achieving profitability

This approach allows transaction volume to scale without a proportional increase in fixed costs, improving overall financial efficiency.

8.1 Margin Expansion Strategy

While the initial model relies heavily on volume, long-term sustainability requires gradual improvement in margin per order.

This section outlines how CampusCart increases profitability without compromising user affordability.

First, commission rates are selectively adjusted. The baseline commission remains at 10%, but premium sellers or high-demand products are subject to higher rates between 12% and 15%. This increases revenue per transaction without affecting general pricing perception.

Second, delivery efficiency improves as order density increases. Through batch delivery, multiple orders within the same area are grouped into a single route. This reduces cost per delivery.

For example, if delivery cost decreases from 8,000 VND to 6,000–7,000 VND, the net delivery margin increases from 2,000 VND to approximately 3,000–4,000 VND per order.

Third, additional revenue streams are introduced. These include promoted listings, vendor subscription plans, and targeted in-app advertising. Unlike transaction-based revenue, these sources do not depend directly on order volume.

Over time, these improvements increase the contribution margin per order. As a result, the platform becomes less dependent on extreme transaction volume to achieve profitability.

In summary, the initial phase prioritizes volume and market penetration, while later stages focus on improving unit economics and revenue diversification.

9 Marketing Strategy

9.1 Market Positioning

CampusCart is positioned as a specialized e-commerce platform tailored specifically for university students. Unlike general-purpose marketplaces, the platform focuses on addressing the unique purchasing behavior of this segment, which prioritizes affordability, convenience, and fast delivery.

By narrowing its focus to a clearly defined user group, CampusCart is able to deliver more relevant offerings and a more streamlined user experience. This targeted approach strengthens differentiation and allows the platform to compete effectively against larger but less specialized competitors.

9.2 Marketing Channels

To effectively reach its target audience, CampusCart leverages a combination of digital and campus-based channels. Social media platforms such as TikTok and Facebook play a central role, as students are highly active and responsive on these platforms.

In addition, campus ambassador programs are used to promote the platform directly within universities, creating a strong peer-to-peer influence. Referral systems further amplify this effect by encouraging users to invite friends, thereby accelerating organic

growth.

This multi-channel strategy reduces customer acquisition cost through peer-driven distribution and referral-based growth. To achieve the required transaction volume, CampusCart adopts a multi-campus expansion strategy. Each campus operates as a localized demand cluster, allowing the platform to replicate its model while maintaining operational efficiency. As the number of campuses increases, total transaction volume scales linearly without requiring proportional increases in fixed costs.

9.3 Performance Measurement

The effectiveness of marketing strategies will be measured using key metrics such as customer acquisition cost, conversion rate, and retention rate. These indicators provide a clear view of campaign performance and support continuous optimization.

9.4 Customer Acquisition

Customer acquisition efforts focus on building an initial user base and generating early traction. Promotional incentives, such as discounts for first-time users, are used to lower entry barriers and encourage trial.

Referral programs are implemented to leverage network effects, allowing existing users to contribute to platform growth. In parallel, campus-based campaigns and partnerships help establish visibility and credibility within the student community.

These strategies reduce customer acquisition cost by leveraging existing social networks within campus environments.

9.5 Customer Retention

Maintaining long-term user engagement is a key priority for CampusCart. Retention strategies focus on encouraging repeat purchases by delivering more relevant product suggestions based on past activity.

Personalized recommendations, powered by user data, help improve product relevance and reduce search effort. Loyalty programs incentivize continued usage, while group buying features increase repeat purchase frequency through shared incentives and collaborative purchasing.

By strengthening retention, the platform increases purchase frequency and customer lifetime value, contributing to sustainable revenue growth.

10 Development Plan

10.1 Phases

The development of CampusCart follows a structured sequence to ensure both technical readiness and market validation.

The first phase focuses on planning, including requirement definition and system design. This is followed by the development phase, where core features such as ordering, payment integration, and vendor management are implemented.

The third phase involves testing, where system reliability and usability are evaluated. Finally, the deployment phase introduces the platform to real users within a controlled campus environment.

This staged approach reduces implementation risk and allows early feedback to inform subsequent improvements.

10.2 Timeline

The initial version of the platform is designed to be completed within four weeks.

The first week is allocated to planning and architecture design. The next two weeks focus on core development, including frontend and backend integration. The final week is dedicated to testing and deployment.

This timeline prioritizes rapid release of a minimum viable product, enabling early validation of user demand and operational assumptions.

10.3 Cost Estimation

CampusCart is designed to operate with a relatively low initial cost structure.

Initial development costs are estimated at 10–15 million VND. Monthly operational costs consist of hosting (1–2 million VND), maintenance (2–3 million VND), and marketing expenses (5–10 million VND).

These components result in an estimated fixed monthly cost of approximately 20 million VND, which is consistent with the break-even analysis presented earlier.

This cost structure allows the platform to scale without requiring significant upfront investment.

10.4 Key Performance Indicators (KPIs)

This section translates the revenue model and unit economics into measurable operational targets.

In the initial phase, CampusCart targets:

- 500–1,000 registered users

- Conversion rate of 10–15%
- Monthly order volume of 900–2,400 transactions

Based on the revenue model:

$$\text{Revenue} = 7,000 \times \text{orders}$$

Projected monthly revenue is:

$$\text{6.3 million VND to 16.8 million VND}$$

Using the contribution margin:

$$\text{Contribution} = 1,000 \times \text{orders}$$

Projected contribution margin is:

$$\text{0.9 million VND to 2.4 million VND}$$

When compared with fixed monthly costs of 20 million VND, the platform operates at a deficit of approximately:

$$\text{17.6 million VND to 19.1 million VND}$$

This result is consistent with the break-even requirement of 20,000 orders per month. Based on these projections, the platform is expected to operate at a net loss during the initial phase, with profitability achieved only when monthly order volume approaches or exceeds the break-even threshold.

10.5 Growth Gap Analysis

To reach break-even, the platform must increase transaction volume significantly.

$$\text{Required scale increase} = \frac{20,000}{900 \text{ to } 2,400} \approx 8.3\times \text{ to } 22.2\times$$

This gap highlights that early-stage performance is not designed for profitability, but for validation.

In practical terms, a single campus cannot generate sufficient volume. Growth must be achieved through geographic expansion and increased order density.

10.6 Scaling Roadmap

A structured scaling strategy is required to close the gap between current performance and break-even.

- **Phase 1 (1 campus)**: 900–2,400 orders/month. Focus on validating demand and user behavior.
- **Phase 2 (3–5 campuses)**: 3,000–10,000 orders/month. Improved vendor network and early delivery optimization.
- **Phase 3 (8–10 campuses)**: 10,000–20,000 orders/month. Approaching break-even, with improved operational efficiency.
- **Phase 4 (10+ campuses)**: >20,000 orders/month. Consistent profitability supported by higher margin and additional revenue streams.

This progression ensures that growth in transaction volume is matched by improvements in operational efficiency.

10.7 Performance Measurement

To ensure continuous alignment between strategy and execution, performance is monitored through several key indicators:

- Customer Acquisition Cost (CAC), to measure marketing efficiency
- Conversion Rate, to evaluate user behavior within the platform
- Retention Rate, to assess repeat purchase patterns
- Orders per User, to track engagement intensity

These metrics are directly connected to revenue generation and are used to guide strategic adjustments.